

ANANYA SAKHALKAR

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 2+ years of experience spanning applied research and production ML systems across healthcare, industrial, agritech, and NLP domains. Published IEEE researcher with hands-on experience fine-tuning Transformer-based LLMs (Llama 3.1) and encoder models (DistilBERT) using both full fine-tuning and PEFT/LoRA within the Hugging Face ecosystem. Strong foundation in deep learning, model evaluation, and reproducible experimentation, combined with a proven ability to independently deliver scalable ML solutions, from raw data pipelines through live inference serving.

TECHNICAL SKILLS

- **LLM / VLM & Transformers:** Llama 3.1, DistilBERT, PyTorch, Hugging Face Transformers, PEFT/LoRA fine-tuning, Transformer architectures, CNNs (ResNet, DenseNet)
- **ML Frameworks:** TensorFlow, Keras, Scikit-Learn, XGBoost, Random Forest
- **Data Preprocessing:** Feature engineering, missing value handling, outlier detection, feature selection, data cleaning & transformation
- **Model Development:** Algorithm selection, model evaluation metrics, performance tuning, experiment tracking, model interpretability
- **Deployment & Infra:** FastAPI, Flask, Docker, REST APIs, cloud deployment, low-latency inference, end-to-end ML pipelines
- **Languages:** Python, C++, SQL
- **Data & Databases:** MongoDB, MySQL, Google Earth Engine, Sentinel-2 remote sensing, OpenCV

PUBLICATIONS

- S. Raman, A. Sakhalkar, and P. Debnath, "Farm Digital Twin and Deep Learning-Based Crop Health Detection Using Multi-Index Vegetation Signals from Sentinel-2 Satellite Imagery," 2026 International Conference on System, Computation, Automation and Networking (ICSCAN), Puducherry, India, 2026. DOI: [10.1109/ICSCAN66520.2026.11588351](https://doi.org/10.1109/ICSCAN66520.2026.11588351).

EXPERIENCE

ML & Data Science Intern · Jio Platforms Limited

Jan 2026 – Apr 2026

- Engineered a real-time predictive analytics system detecting site-level failures across a nationwide telecom network — processing live sensor streams (RSRP, SINR, throughput) with sub-second latency.
- Developed and tuned TensorFlow/Scikit-Learn models for network degradation prediction, enabling proactive intervention before failures escalated and reducing potential service downtime.
- Designed Python-based anomaly detection algorithms integrated into automated decision pipelines, improving operational response time by 20–30%.
- Delivered end-to-end data preprocessing and feature engineering pipelines with visualization and forecasting layers, translating raw telemetry into actionable insights.

ML Engineer Intern · APMSIDC – Govt. Healthcare Platform

Feb 2025 – May 2025

- Built and shipped an end-to-end TensorFlow/Scikit-Learn inference pipeline for a precision-medicine recommendation system on a live government healthcare platform serving real patients.
- Achieved <200ms response time by architecting an optimized FastAPI inference service, meeting strict production SLA requirements for real-time clinical decision support.
- Designed comprehensive feature engineering and preprocessing workflows for sensitive clinical data, including missing value handling, outlier detection, and feature selection.
- Applied model interpretability techniques and established model evaluation workflows, systematically improving recommendation accuracy over multiple iterations.

Software Engineering Intern · Reliance Industries Limited

May 2024 – Jul 2024

- Developed and deployed ML solutions for biotech and industrial use cases in a production-grade environment; presented results directly to internal research leadership.

- Delivered ~90%+ classification accuracy and 20–30% improvement in anomaly detection speed across the full model lifecycle: design, training, evaluation, debugging, and iteration.
- Executed comprehensive data preprocessing pipelines and performance tuning to meet business-critical accuracy and efficiency requirements.

Freelance AI/ML Developer · Independent

2023 – Present

- Sole architect and engineer across full product lifecycles independently designing, training, deploying, and iterating on ML systems delivered to clients.
- Built a desktop AI-powered Python IDE integrating ML models for intelligent code suggestions, error detection, and automated developer assistance.

PROJECTS

Mental Health Support Chatbot & Emotion Classification System | *DistilBERT · Llama 3.1 8B Instruct · Hugging Face Transformers · PEFT/LoRA*

- Full fine-tuned DistilBERT on public Hugging Face and Kaggle datasets to classify user messages across 8 emotional-state categories — Anxiety, Depression, Stress, Loneliness, Anger, Happy/Positive, Neutral, and Crisis Risk — achieving 93.4% accuracy, 92.8% precision, 92.1% recall, and 92.4% F1-score.
- Fine-tuned Llama 3.1 8B Instruct using LoRA (PEFT) to generate emotionally-aware chatbot responses conditioned on the predicted emotional state, reducing training cost versus full-parameter fine-tuning while preserving response quality.
- Built end-to-end with the Hugging Face Transformers ecosystem, covering dataset curation, tokenization, training, and evaluation in a reproducible pipeline.

Farm Digital Twin — Crop Simulation & Health Detection | *Published Research, IEEE ICSCAN 2026 · Sentinel-2 · Google Earth Engine · Random Forest · CNN*

- Co-authored a peer-reviewed paper (IEEE ICSCAN 2026) on an integrated crop health monitoring system combining satellite remote sensing, statistical anomaly detection (Z-score normalization), and ML classification.
- Architected a digital twin using Sentinel-2 satellite imagery and Google Earth Engine to compute vegetation indices (NDVI, EVI, FPAR, SVI) and model crop health dynamics across geographic regions.
- Achieved ~96% classification accuracy distinguishing healthy vs. stressed/diseased crop regions using Random Forest and CNN classifiers trained on satellite-derived features.

Drug Recommendation System — Pharmacogenomics | *XGBoost · Random Forest · Neural Networks · Scikit-Learn*

- Built a multi-model recommendation engine for gene–drug interaction prediction, achieving ~85–90% accuracy across held-out clinical datasets using robust preprocessing and model evaluation pipelines.
- Applied interpretability tools to validate recommendations against domain knowledge; implemented a full feature engineering pipeline including outlier detection and feature selection.

Plant Disease Detection | *ResNet · DenseNet · TensorFlow · Keras · OpenCV*

- Trained and tuned CNN models (ResNet, DenseNet) using TensorFlow/Keras for multi-class crop disease classification with extensive data augmentation to handle real-world variability.
- Deployed a cloud-based inference system with integrated treatment recommendations — full end-to-end production deployment.

EDUCATION

B.Tech in Computer Science · SRM Institute of Science and Technology, Chennai

CGPA: 8.94/10 | 2022 – 2026

CERTIFICATIONS

- AWS Academy: Machine Learning Foundations (2024)
- Cisco: Introduction to Data Science (2025) · Introduction to Modern AI (2025)
- NPTEL: Cloud Computing (2025)